

# Intrusion Detection Using MLP and Autoencoder-Based Feature Extraction on the CICIDS2017 Dataset

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**Abstract**—Intrusion detection systems play an important role in identifying malicious network activities and protecting modern computer networks. This paper evaluates two deep learning-based approaches for multiclass intrusion detection using the CICIDS2017 dataset. The first approach uses a Multilayer Perceptron (MLP) classifier trained directly on selected traffic features, while the second approach uses an Autoencoder as a feature extractor followed by an MLP classifier. Experimental results show that both models achieve high overall performance on dominant traffic classes. However, both models are affected by severe class imbalance, especially for rare attack categories. Although the Autoencoder + MLP model achieves higher weighted precision, recall, F1-score, ROC-AUC, and PR-AUC, its balanced accuracy is lower than the standalone MLP model, indicating poor minority-class detection. Therefore, the results demonstrate that high overall accuracy alone is not sufficient for evaluating intrusion detection models on imbalanced datasets.

**Index Terms**—Intrusion Detection System, CICIDS2017, MLP, Autoencoder, Deep Learning, Class Imbalance, Network Security

## I. INTRODUCTION

With the increasing number of cyber threats, intrusion detection systems (IDSs) have become essential for monitoring network traffic and identifying malicious activities. Machine learning and deep learning methods are widely used in IDS research because they can learn complex traffic patterns from large datasets. However, real-world intrusion detection datasets are often highly imbalanced, where normal traffic and common attacks dominate the dataset while rare attacks have very few samples.

This study compares two models for multiclass intrusion detection: a standalone MLP classifier and an Autoencoder-based feature extraction model followed by an MLP classifier. The main objective is to evaluate not only the overall performance but also the ability of each model to detect minority attack classes.

## II. DATASET DESCRIPTION

The experiments were conducted using the CICIDS2017 dataset. The dataset contains network traffic records belonging to 15 classes, including BENIGN traffic and multiple attack categories such as DDoS, DoS Hulk, PortScan, Bot, Heartbleed, Infiltration, and Web Attack variants. The dataset is highly imbalanced, with BENIGN traffic representing the

majority class and several attack classes containing very few samples. For example, rare classes such as Heartbleed, Web Attack - SQL Injection, and Infiltration contain very limited instances compared with BENIGN traffic and common attacks.

## III. METHODOLOGY

### A. Feature Selection and Preprocessing

A total of 40 selected network traffic features were used for training and evaluation. The features include packet length statistics, flow duration, inter-arrival time features, header length features, packet rate features, and byte-based flow features. The label column was encoded using label encoding, and feature values were standardized using StandardScaler before training the models.

### B. MLP Model

The first model is a standalone MLP classifier. It uses the selected 40 features directly as input. The MLP architecture consists of three hidden layers with 256, 128, and 64 neurons respectively, using the ReLU activation function. The Adam optimizer was used for training, and early stopping was applied to reduce overfitting.

### C. Autoencoder + MLP Model

The second model uses an Autoencoder for feature extraction. The encoder compresses the 40-dimensional input features into a lower-dimensional latent representation. The encoder layers contain 32, 16, and 8 neurons. The decoder reconstructs the original input from the latent representation. After training the Autoencoder using reconstruction loss, the encoder output is used as input to an MLP classifier for final multiclass classification.

## IV. EVALUATION METRICS

The models were evaluated using precision, recall, F1-score, PR-AUC, ROC-AUC, balanced accuracy, Matthews Correlation Coefficient (MCC), specificity, and G-Mean. Weighted metrics were used to measure overall performance, while macro and balanced metrics were used to evaluate fairness across all classes. Since the dataset is highly imbalanced, balanced accuracy, PR-AUC, and per-class metrics are especially important for understanding minority-class performance.

TABLE I  
PERFORMANCE COMPARISON OF MLP AND AUTOENCODER + MLP  
MODELS

| Metric                   | MLP          | Autoencoder + MLP |
|--------------------------|--------------|-------------------|
| Precision (%)            | 76.12        | <b>97.20</b>      |
| Recall (%)               | 62.76        | <b>97.39</b>      |
| F1 Score (%)             | 65.40        | <b>97.23</b>      |
| Balanced Accuracy (%)    | <b>62.76</b> | 49.72             |
| ROC-AUC (%)              | 97.83        | <b>99.64</b>      |
| PR-AUC (%)               | 66.98        | <b>99.40</b>      |
| MCC (%)                  | <b>97.63</b> | 91.09             |
| Specificity (%)          | <b>99.78</b> | 90.03             |
| G-Mean (%)               | 68.04        | <b>93.64</b>      |
| Minority Class Detection | Weak         | Very Poor         |
| Model Complexity         | Low          | High              |
| Reliability              | Moderate     | Low               |

## V. RESULTS AND DISCUSSION

### A. Performance Comparison

Table I presents the comparison between the standalone MLP model and the Autoencoder + MLP model.

The results show that the Autoencoder + MLP model achieves higher weighted precision, recall, F1-score, ROC-AUC, and PR-AUC compared with the standalone MLP model. However, its balanced accuracy is only 49.72%, which is lower than the MLP model's balanced accuracy of 62.76%. This indicates that although the Autoencoder + MLP model performs well overall, it performs poorly when all classes are treated equally.

### B. MLP Model Analysis

The standalone MLP model achieved a macro recall of 62.76%, macro precision of 76.12%, and macro F1-score of 65.40%. These results indicate moderate performance across all classes. The model achieved a high ROC-AUC score of 97.83%, showing strong class separability. However, the PR-AUC score of 66.98% and G-Mean of 68.04% reveal that the model still struggles with minority classes. The weighted F1-score appears very high because it is strongly influenced by the majority classes.

### C. Autoencoder + MLP Model Analysis

The Autoencoder + MLP model achieved weighted precision of 97.20%, weighted recall of 97.39%, and weighted F1-score of 97.23%. These values suggest excellent overall performance. However, the balanced accuracy of 49.72% shows that the model performs poorly across classes when class imbalance is considered. Several minority classes such as Bot, Heartbleed, Infiltration, Web Attack - Brute Force, Web Attack - SQL Injection, and Web Attack - XSS achieved zero precision, recall, and F1-score. This confirms that the model is highly biased toward majority classes.

## VI. GRAPH ANALYSIS

### A. ROC Curve

The ROC curve shows that the models have strong class separability, with most classes achieving high AUC values. For the Autoencoder + MLP model, the ROC-AUC reached 99.64%, suggesting that the learned encoded features are highly separable. However, ROC curves can be overly optimistic for imbalanced datasets because false positive rates remain low when the negative class is very large. Therefore, the ROC curve does not fully reflect poor minority-class detection.

### B. Precision-Recall Curve

The Precision-Recall curve provides a more realistic evaluation for this dataset. It shows that dominant classes such as BENIGN, DDoS, and DoS Hulk achieve strong performance, while minority classes show very low precision and recall. This indicates that the models struggle to detect rare attacks reliably. Therefore, the Precision-Recall curve is more informative than the ROC curve for this imbalanced intrusion detection problem.

### C. Confusion Matrix

The confusion matrix shows strong correct predictions for majority classes along the diagonal. However, many minority-class samples are misclassified as BENIGN or other dominant classes. This confirms that the models are biased toward common traffic patterns and fail to detect rare but critical attacks. In a real-world IDS, this is a serious limitation because rare attacks can still cause significant security damage.

### D. Autoencoder Loss Curve

The Autoencoder loss curve shows a steady decrease in both training and validation loss. This indicates that the Autoencoder successfully learns meaningful compressed representations of the input features. The close trend between training and validation loss suggests that there is no major overfitting during feature learning.

### E. Classifier Loss Curve

The classifier loss curve shows stable convergence, with both training and validation loss decreasing during training. This indicates that the classifier learns effectively from the encoded features. However, stable training loss does not guarantee strong minority-class detection, as shown by the per-class results and confusion matrix.

## VII. LIMITATIONS

The main limitation of both models is the severe class imbalance in the CICIDS2017 dataset. Majority classes dominate the evaluation metrics, causing weighted metrics to appear very high even when rare attack classes are not detected. Another limitation is that the Autoencoder compresses the input features without using class information, which may remove useful minority-class information during feature compression.

## VIII. CONCLUSION

This study compared a standalone MLP classifier and an Autoencoder + MLP model for multiclass intrusion detection using the CICIDS2017 dataset. The Autoencoder + MLP model achieved higher overall weighted metrics, but its lower balanced accuracy and poor minority-class performance show that it is less reliable for real-world intrusion detection. The standalone MLP model achieved lower overall metrics but performed better in terms of balanced accuracy. Overall, the results show that high weighted accuracy, ROC-AUC, and PR-AUC are not sufficient for evaluating IDS models on imbalanced datasets. Future work should apply class balancing techniques such as SMOTE, class weighting, focal loss, or cost-sensitive learning to improve minority attack detection.

## FUTURE WORK

Future research can improve the detection of minority attack classes by applying oversampling methods such as SMOTE or ADASYN, using class-weighted loss functions, applying focal loss, and evaluating tree-based models such as XGBoost and LightGBM. Additional experiments can also include feature importance analysis and cross-validation to improve model robustness.

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